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SepConv-ens: An ensemble of separable convolution-based deep learning models for weather radar echo temporal extrapolation

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Abstract

The paper addresses the topic of radar echo temporal extrapolation which is of major interest in both operational and research meteorology. Weather radar measurements are an important data source used by operational meteorologists for weather analysis, radar reflectivity having a significant influence on short-term heavy rainfall prediction. Thus, extrapolating radar products' values is important for early storm evolution assessment. The paper proposes SepConv-ens approach for temporal extrapolation of radar observations using an ensemble of three separable convolution-based deep learning models. Experiments performed on real radar data from the Romanian National Meteorological Administration (NMA) highlight a good performance of SepConv-ens in predicting radar data up to more than 40 minutes ahead and a good correlation between the radar measurements and the predictions in terms of spatial and intensity evolution of the radar echoes. SepConv-ens is integrated in the operational visualization software utilized by the Romanian NMA and is the first attempt, at the national level, to offer an artificial intelligence-based automated assistance for operational meteorologists.

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1. Introduction

Both conventional *machine learning* (ML) and *deep learning* (DL) techniques are increasingly used in the scientific literature for improving meteorological tasks such as uncovering relevant patterns regarding the weather evolution quantitative precipitation estimation, rainfall or hailstorm prediction DL models such as *convolutional neural networks* (CNN), *generative adversarial networks* (GANs) [17], *recurrent neural networks* (RNNs) [11] became popular models applied on radar or satellite data to automatically extract relevant features useful in various meteorological tasks such as weather nowcasting. Data obtained from weather radars is an important data source used by nowcasting

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meteorologists for assessing current weather and to provide very-short term forecasts, up to 90 minutes lead-time. Reflectivity (R) is one of the main radar products used in operational environments for monitoring the weather evolution (e.g., spatio-temporal evolution of precipitating clouds). Despite its practical importance for weather nowcasting and issuing nowcasting warnings, extrapolating the radar echoes in operational environments strongly depends on the skills and experience of the meteorologist and thus presents weaknesses in terms of processing times and precision. DL-based techniques recently employed for radar echo extrapolation surpass the traditional methods such as cross-correlation, individual radar echo-tracking or the optical flow method [4] in terms of accuracy and efficiency. However, certain challenges such as the high-dimensional and heterogeneous character of weather data, and the relative scarcity of severe weather events, which are of interest for nowcasting, still make it difficult for ML models to precisely predict the occurrence of extreme events.

The work conducted in this paper is in line with the research conducted in the literature on defining new DL-based architectures for radar echo temporal extrapolation. In this regard, the paper proposes SepConv-ens approach for extrapolation of radar observations up to one hour ahead using an ensemble of three separable convolution-based DL models. Considering the interval between two radar acquisitions as a time step (6 minutes in our case study), we are employing three base DL models trained to predict radar data for 1, 3 and respectively 5 time steps ahead using data from four previous time steps as input. For enhancing the performance, the prediction for multiple time steps ahead is achieved by combining the predictions of the base DL models. Real radar data from the NMA will be employed for evaluating the predictive performance of SepConv-ens using well-known evaluation metrics employed in the forecasting literature. An interpretation of the results from a meteorological perspective is additionally provided. The proposed SepConv-ens model was integrated in the operational visualization software utilized by the Romanian NMA and validated from an operational perspective as well. To the best of our knowledge, the approach introduced in this paper is new in the literature, while being the first attempt, at the national level, to offer an artificial intelligence-based automated assistance for operational meteorologists in short-term weather forecasting.

To summarize, our study aims to answer to the following research questions: **RQ1** How to design an ensemble of separable convolution-based DL models for temporal prediction of radar echoes up to one hour ahead and how does it perform on real radar data provided by the NMA and collected from the Central Romania region? **RQ2** Would such an ensemble model be more efficient than the base models? How does the extrapolation accuracy compare to similar related work? **RQ3** How effective would be the proposed model in an operational environment as a tool for assisting operational meteorologists in decision-making related to high impact weather?

The rest of the paper is organized as follows. Section 2 reviews recent similar existing work, related to radar data extrapolation using DL models. Section 3 presents the methodology employed for designing and validating the SepConv-ens ensemble model and presents the case study used. Section 4 presents our experimental results and discusses on the model's performance and its current limitations from a meteorological perspective. The last section outlines the conclusions and presents directions to further improve the current work.

2. Literature review

We summarize, in the following, recent DL-based radar data extrapolation techniques. By highlighting various literature approaches, we aim to extract the general research directions currently under investigation. The exploration of particular optimisations applied on previously proposed architectures constitutes one such major direction of research. Many authors focus on proposing individualistic solutions for the challenges encountered during the original development of DL architectures. For example, to tackle the issue of error accumulation given by exclusively using the mean square error and mean absolute error, a new instance-based loss has been introduced [15], a loss function that is dynamically adjusted to the pre-processed ground truth data. The authors demonstrate the effectiveness of using such a loss function by training four DL architectures: the ConvLSTM, ConvGRU, PredRNN and PredRNN++ networks. In all experiments, the models trained using the proposed loss functions manifest performance improvements when compared to the original training process. Other approaches, such as applying the SSIM and MS-SSIM [14] loss functions, introducing a Plug-and-Play attention-based module [12] or even improving the data preprocessing by applying noise and ground clutter filtering [16] share a similar objective: to improve the performance of existing DL-based radar data extrapolation techniques.

By contrast, proposing and defining new DL-based architectures comprise another major research direction. This research direction is tightly coupled with the general trends ongoing in the Computer Vision research space. We'll

start by presenting recent CNN-based advances in radar echo extrapolation. In order to take into account the impact of the environmental factor, a 3D-CNN has been devised [13] to incorporate multiple data sources across the spatiotemporal dimension. The technique works by applying a late fusion strategy, enabling the model to correlate radar echo images with environmental data during training. Other advanced CNN-specific methods have also been shown to provide promising results. Specifically, combining dilated and attention convolutions [9] allows the model to preserve the internal data structure of the feature matrix, leading to performance gains.

Another set of approaches involves the usage of GAN models as core foundations for building radar echo extrapolation architectures. One such method [16] adapts the Wasserstein GAN framework by integrating convolutional recurrent units with a gate mechanism to essentially capture and preserve the spatiotemporal information. To address the issue of echo attenuation, where the correlation between real rainfall evolution and generated images weakens as the lead time increases, the architecture introduced in [16] has been extended [17]. The new architecture explicitly defines a spatiotemporal image correlation prediction network, which uses attention mechanisms to suppress the blurring effect of rain distribution and reduce the negative bias. Another GAN-based approach involves training a spatiotemporal differential information extractor [7] to explicitly capture the temporal variation across image frames. This approach has demonstrated strong performance on generalizing up to 2 hours in the future, despite being trained to predict only 1 hour in advance.

Last but not least, Vision Transformers foundational models are finding their place in the field of radar echo extrapolation. *TempEE* [5] is a Temporal-Spatial Parallel Transformer which makes use of Multi-level Spatial-Temporal Attention to capture global and local information in an efficient manner. Not only it avoids using auto-regression, but the method also employs a parallel encoder to extract representations about the continuous motion process of the echo images. By successfully preserving the advantages of Transformer-based techniques demonstrated in Computer Vision, *TempEE* stands as the current state-of-the-art method in radar echo extrapolation.

Empirical observations across the related work on radar data temporal extrapolation revealed that a *critical success index*, *probability of detection* and *Heidke Skill score* range of [0.5, 0.7] coupled with a *false alarm ratio* range of [0.1, 0.4] generally denote a stable and performant model.

3. Methodology

This section answers research question **RQ1** by introducing the methodology employed for developing and evaluating SepConv-ens model designed for predicting the values of radar products up to k time steps ahead based on the currently available radar data (one time step is considered the time between two consecutive radar acquisitions).

Our study is conducted on real radar data provided by the Romanian NMA. The data was collected by a Doppler single-polarization weather radar located in central Romania. About every 6 minutes the radar outputs approximately 30 base and derived products, related to the location, intensity and the movement of water particles within a specific radius (i.e., 230 km) around the radar. In our experiments we will use the base Reflectivity (R) product providing information on location, intensity and type of precipitation, storm structure and size and distribution of water particles. The reflectivity product is available for several elevation angles of the radar antenna, and for each time step a set of six products (R01-R04 and R06-R07) is delivered, each of them corresponding to a certain tilt of the antenna. The data set employed in our study contains 211 days, manually curated by meteorologists, with convective events collected from 2020, 2021 and 2022. The events occurred in central Romania, on an area defined by a 230 km radius relative to the radar site. While the data set is focused on convective events, it contains, to a lesser extent, a number of stratiform events. The geographical area is visualized as an $n \times m$ grid, where each point (x, y) ($1 \leq x \leq n$, $1 \leq y \leq m$) represents a geographical location: x corresponds to the longitude of the location and y corresponds to the latitude of the location. For our data both m and n are 460. Figure 1 presents the distribution of R01 values in the considered data set, the OX axis denoting the time stamps (240 radar acquisitions for each day from the data set). For any acquisition we computed the number of values in the ranges [20,30), [30,40) and [40, 70). A single radar acquisition contains 460×460 data points so a value of 20000 for [30,40) means that only around 10% of the data points have R01 values ranging in the interval [30,40). Even if the data was specially selected by meteorologists to contain days with convective events, the data is highly imbalanced and high reflectivity values represent only around 4% of the total measurements.

3.1. Formalisation, data representation and preprocessing

Let us consider that Rp is a set of radar products of interest $Rp = \{r_1, r_2, \dots, r_p\}$. The raw radar data from the analyzed geographical area, at time t , is visualized as an 3D image with p -channels: the OX axis represents the

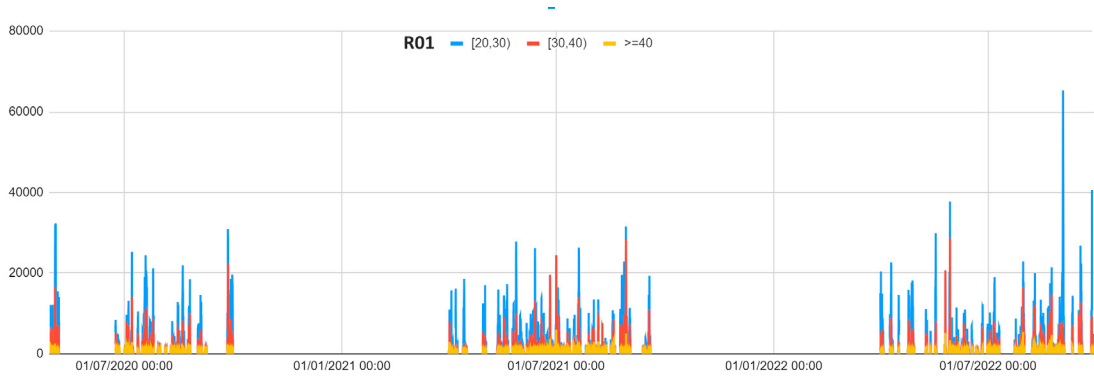


Fig. 1: Distribution of R01 values in all radar acquisitions from the data set (211 days with convective events, with 240 acquisitions per day).

longitude value of the geographic map, the OY axis represents the latitude value of the area and the OZ axis represents the radar products values. More specifically, $M_l(n, m, p) = (m_{ijk}^t)_{i=\overline{1,n}, j=\overline{1,m}, k=\overline{1,p}}$, where $n \times m$ is the dimensionality of the data grid and m_{ijk}^t stores the value of the radar product r_k at time stamp t , in the location (i, j) from the grid. Assuming that l is a predefined number of time steps, our learning task is to predict the sequence of 3D images at time moments $t + 1, \dots, t + k, k > 1$ (i.e., $6 \cdot k$ minutes ahead) from a sequence of l previously collected 3D radar images (i.e., the data collected at $t - l + 1, \dots, t$). We note that one time step (the period between two consecutive time moments $t - 1$ and t) is about 6 minutes (the time between two radar acquisitions). Thus, the target function to be learned is formalized as a function Φ that maps an l -length sequence of p -channel 3D images (representing the radar data at $t - l + 1, \dots, t$) to an k -length sequence of p -channel 3D images (corresponding to the images at $t + 1, \dots, t + k$), i.e., $\Phi(M_{t-l+1}, \dots, M_t) = \langle M_{t+1}, \dots, M_{t+k} \rangle$.

Our proposed SepConv-ens deep learning model will be designed to learn a hypothesis $\widehat{\Phi}$ that is an approximation of the target function Φ ($\widehat{\Phi} \approx \Phi$) such that $\widehat{\Phi}(M_{t-l+1}, \dots, M_t) = \langle \widehat{M}_{t+1}, \dots, \widehat{M}_{t+k} \rangle$, where $\widehat{M}_{t+i}$ denotes the predicted 3D image at time stamp $t + i$ that approximates the real 3D radar image at $t + i$ at a certain confidence level, i.e., $\widehat{M}_{t+i} \approx M_{t+i} \forall 1 \leq i \leq k$. Thus, for a sequence of radar images collected at $t - l + 1, \dots, t$, SepConv-ens will provide a sequence of multi-channel 3D images representing the estimation of radar images at time moments $t + 1, \dots, t + k$.

A data set $\mathcal{D}$ formed from sequences $\langle M_{t-l+1}, \dots, M_t \rangle$ of 3D images collected on various days with convective events together with their ground truth labels is available and will be further used for building SepConv-ens. The ground truth label for an input sequence $\langle M_{t-l+1}, \dots, M_t \rangle$ is a k -length sequence $\langle M_{t+1}, \dots, M_{t+k} \rangle$ of p -channel radar images. In our case study, we are using 6 radar products (i.e., $p = 6$ and $Rp = \{R01, R02, R03, R04, R06, R07\}$). Thus, the radar data set $\mathcal{D}$ used for training consists of radar reflectivity sampled at 6 elevation angles, while the learning model output consists in radar reflectivity values estimated for k time steps ahead and computed from all 6 elevations, on a geographic area defined by a 230 km radius relative to the radar site. To predict the reflectivity values up to a given time interval ahead, the model uses the radar reflectivity from l previous time steps as input. Because of different factors, such as variation in time step length or shutdowns of the radar, in our data set, the number of time steps per day can vary between 230 to 266, with an average of 250 time steps per day. As a data preprocessing step, the missing values are replaced by 0 and then all values are scaled to $[0, 1]$ using the min-max normalization method. Regarding corrupted data, which could result from sensor errors or some geographic features such as high mountains, this is corrected based on known data patterns, for example by removing the ground clutter.

3.2. The proposed SepConv-ens model

We consider, in the following, as a proof of concept, a particularization of the theoretical model from Section 3.1. We are using 4 previous time steps as input for the models (i.e., $l = 4$), while the prediction is made for 8 steps ahead (i.e., $k = 8$ – about 48 minutes ahead). Based on meteorological feedback, using 4 previous time steps is justified as an empirical determined optimal balance between capturing the essential dynamics of convective storm events, while ensuring computational efficiency and minimizing model complexity. The 4 previous steps represent about 25 minutes and as the lifespan of convective events ranges from about 30 minutes, for single-cells, to over 12 hours for

large mesoscale convective systems, the selected interval should capture the dynamic of the events. The proposed SepConv-ens model consists of an ensemble of three separable convolution-based DL models. Depthwise separable convolutions are used instead of normal convolutions because they highly increase the computational efficiency of the neural network. For modelling the temporal evolution of the radar data but in the meantime for reducing the computational resources and speeding up the training process we decided to employ non-recurrent neural networks and input data from multiple time steps.

The three base DL models employed in the SepConv-ens ensemble (further denoted by Bm_1 , Bm_3 and Bm_5) are trained to predict the radar data at 1, 3 and respectively 5 time steps ahead using data from four previous time steps as input, i.e., each model Bm_i ($\forall i \in \{1, 3, 5\}$) will be trained to learn an approximation $\widehat{\Phi}_i$ of the function Φ_i such that $\widehat{\Phi}_i(M_{t-3}, M_{t-2}, M_{t-1}, M_t) = \widehat{M}_{t+i} \approx M_{t+i}$. The individual hypotheses $\widehat{\Phi}_i$ learned by the base models will be then combined for obtaining predictions up to 8 time steps ahead. In designing the SepConv-ens ensemble model we started from the assumption that three base models would be sufficient to obtain good enough predictions up to 8 time steps ahead. Additionally, we hypothesise that using a single base model to directly predict the radar data at more than 5 steps (30 minutes) ahead would not be feasible from a meteorological viewpoint, as the rapidly changing character of convective systems leads to high data variability over long intervals, making the training data potentially irrelevant.

At a certain time moment t , SepConv-ens will predict a sequence of p -channel 3D images $\langle \widehat{M}_{t+1}, \dots, \widehat{M}_{t+8} \rangle$ by combining the predictions of the base models Bm_1, Bm_3, Bm_5 and using data available at time steps preceding t as input. Assuming that the predictions of the model Bm_5 may be less accurate than the predictions of Bm_3 , we are using a combination of the base models such that to minimize the use of Bm_5 but employing as much as possible real data in generating the predictions. For predicting the radar data for the first 4 time steps ahead ($t+1, \dots, t+4$), we are applying the models in the order Bm_1, Bm_3, Bm_3, Bm_5 by employing real radar data available at time moments $t-4, t-3, t-2, t-1$ and t . Afterwards, for predicting the radar data at $t+5, \dots, t+8$, the base models are applied in the same order Bm_1, Bm_3, Bm_3, Bm_5 . Thus, the radar data predicted by SepConv-ens are obtained as follows. $\widehat{M}_{t+1}$ is obtained as the prediction of the model Bm_1 applied to the real radar images from four previous time steps, i.e., $\widehat{M}_{t+1} = \widehat{\Phi}_1(M_{t-3}, M_{t-2}, M_{t-1}, M_t)$. Then, $\widehat{M}_{t+2}$ and $\widehat{M}_{t+3}$ are obtained as the predictions of the model Bm_3 applied to the available real radar images, i.e., $\widehat{M}_{t+2} = \widehat{\Phi}_3(M_{t-4}, M_{t-3}, M_{t-2}, M_{t-1})$ and $\widehat{M}_{t+3} = \widehat{\Phi}_3(M_{t-3}, M_{t-2}, M_{t-1}, M_t)$. $\widehat{M}_{t+4}$ is subsequently obtained as the prediction of the model Bm_5 applied to the real radar images at $t-4, t-3, t-2, t-1$, i.e., $\widehat{M}_{t+4} = \widehat{\Phi}_5(M_{t-4}, M_{t-3}, M_{t-2}, M_{t-1})$. The next 4 predictions are obtained by applying Bm_1, Bm_3, Bm_3 , and Bm_5 as follows: $\widehat{M}_{t+5} = \widehat{\Phi}_1(\widehat{M}_{t+1}, \widehat{M}_{t+2}, \widehat{M}_{t+3}, \widehat{M}_{t+4})$, $\widehat{M}_{t+6} = \widehat{\Phi}_3(M_t, \widehat{M}_{t+1}, \widehat{M}_{t+2}, \widehat{M}_{t+3})$, $\widehat{M}_{t+7} = \widehat{\Phi}_3(\widehat{M}_{t+1}, \widehat{M}_{t+2}, \widehat{M}_{t+3}, \widehat{M}_{t+4})$, and $\widehat{M}_{t+8} = \widehat{\Phi}_5(M_t, \widehat{M}_{t+1}, \widehat{M}_{t+2}, \widehat{M}_{t+3})$.

We note that the estimations of the radar data up to 24 minutes ahead ($t+1, \dots, t+4$) is based only on real radar data and thus it is expected that these predictions will have a high enough accuracy. The mechanism to combine the predictions of the base models may be further optimized in order to enhance the predictive performance.

Base architectures. The base models Bm_1 , Bm_3 and Bm_5 , are trained on the same neural network architecture, which can be seen in Figure 2. Each model has 4 inputs corresponding to the input data used by each model for the prediction, as described in Section 3.2. For each input there is a Separable Convolution layer to increase the numbers of filters (or dimensions) from 6 to 51 (the initial 6 filters representing the 6 radar products used – R01-R04 and R06-R07). Then the resulted images are concatenated to create one image with 204 filters. The neural network is continued with 10 Separable Convolution layers, reducing the numbers of filters to 50 in 2 steps and finally there is the output layer which outputs an image with 6 filter - representing the 6 products. After each Separable Convolution layer an activation layer was added with the Scaled Exponential Linear Units (SELU) activation [6]. All the previously mentioned hyperparameters (number of layers, number of filters, activation function) were experimentally determined.

Performance evaluation. The data set is split into 3 parts: testing, validation and training. The testing and validation subsets each contain 10% of the original data while the rest of 80% forms the training subset. All 3 base models use the same training and validation data sets in the training phase (obviously, the input-output pairs used for training will not be the same for the 3 models). For ensuring that the testing set is as representative and reliable as possible, the split into training/validation/testing was done on a day by day basis, even by periods of day: each day was split into four 6-hour periods. The data in each period was split sequentially: the first 10% of the datapoints in the period were assigned to testing, the next 10% were assigned to validation and the rest of 80% were assigned to the training dataset.

For evaluation we are using metrics employed in the *regression*, *classification* and *image processing* literature. The values for the regression and image processing metrics are computed for all points on a radar map as well as only

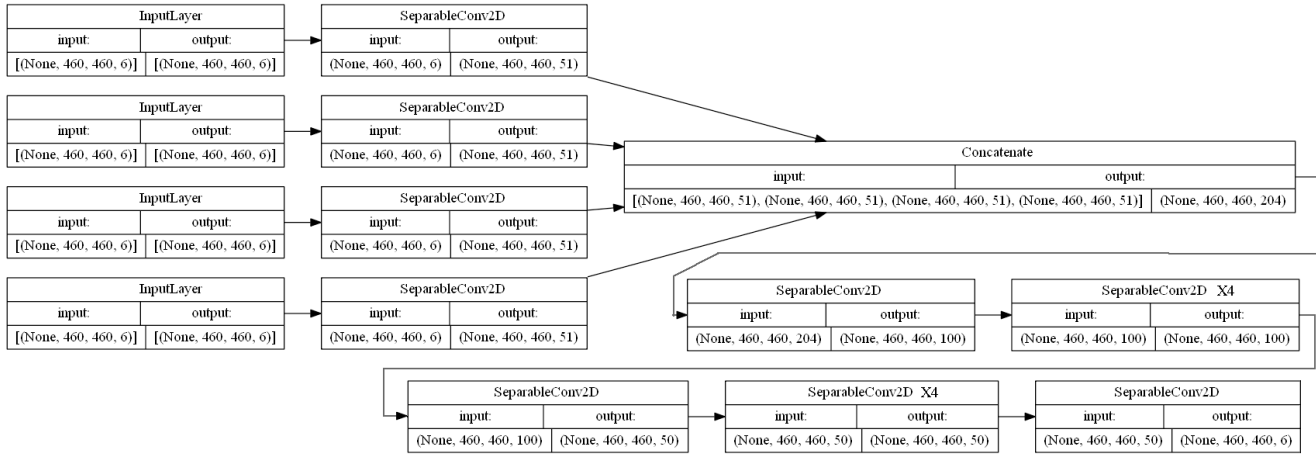


Fig. 2: The architecture of the base neural network models Bm_1 , Bm_3 and Bm_5 . All **SeparableConv2D** layers are followed by an Activation layer with the SELU activation function. "X4" in the name of the layer means that the layer is repeated 4 times.

for the *non-zero* valued points, as zero values are not that interesting – they represent areas where there was no radar echo, meaning that there is clear weather in that area. From a meteorological point of view it is much more interesting to analyse the non-zero valued predictions (e.g., a storm evolution). In addition, there are much more zero values in the training data than non-zero values and thus for the models will be much easier to learn to predict zero values.

Two **regression** metrics will be used in our evaluation: (1) the *Root mean squared error (RMSE)* computed as the average of the RMSE values obtained over the testing samples; and (2) the R^2 score or *coefficient of determination*, a statistical measure of how well the predicted values approximate the real observation (ground truth). For operational meteorologists, not only the values predicted for the radar echoes are relevant, but also the classes of radar products' values. For transforming our regression task into a binary **classification** one, we applied a threshold τ on the continuous output predicted by the DL model. Thus, the values which are higher than τ are considered to be *positive*, while the values which are lower than τ are considered to be *negative*. The ground truth of the instances from the testing data set are also mapped into the *positive* or *negative* class as previously shown. The values from the confusion matrix over the testing data are then computed: TP - the number of positive instances correctly classified as positive; FP - the number of negative instances that were classified as positive; TN - the number of negative points correctly classified as negative; FN - the number of positive instances that were classified as negative. Based on these values, four evaluation metrics usually used in forecasting are computed: *probability of detection* $POD = \frac{TP}{TP+FN}$; *false alarm ratio* $FAR = \frac{FP}{FP+TP}$; *critical success index* $CSI = \frac{TP}{TP+FN+FP}$; and *Heidke Skill Score (HSS)* which measures how well a forecast did relative to a randomly selected forecast, $HSS = \frac{2 \cdot (TP \cdot TN - FP \cdot FN)}{(TP+FN) \cdot (FN+TN) + (TP+FP) \cdot (FP+TN)}$. POD , FAR and CSI measures range in $[0,1]$, while HSS values fall in $[-1, 1]$. For the FAR metric, smaller values are better, while the other measures should be maximised in order to obtain better classifiers. From the **image processing** domain, two evaluation measures are used: VIF (*Visual Information Fidelity*) and F_{cbc} (*Index for the cell-by-cell comparison*). Both metrics are computed for all points and only for the non-zero valued points. VIF is a full reference image quality assessment index that has been proposed by Sheikh and Bovik [8] and quantifies the visual quality of images by capturing the effects of linear contrast enhancements on images. To compute this measure the *sewar* [2] library has been used. Values of VIF greater than 1 indicate an improvement of the visual quality, thus higher values suggest better image quality. F_{cbc} represents the ratio of cells in the predicted image that match with the corresponding cell in the ground truth image. F_{cbc} ranges in $[0,1]$ and have to be maximized for better image predictors.

4. Results and discussion

To complete the answer to **RQ1**, this section presents the performance assessment of SepConv-ens. The models were implemented using the TensorFlow library inside the Keras framework. The models were trained using the gradient descent algorithm with the AMSGrad version of the ADAM optimizer, with a learning rate of 0.001. We also

adjusted the learning rate using a callback, halving the learning rate if the loss value on the validation data sets reached a plateau. The loss function used was a modified Mean Squared Error to account for the fact that there are much fewer high values than lower values (an imbalance spanning several degrees of magnitude) – for this we adjusted the loss to give more weight to higher values. For performance evaluation using classification metrics, the continuous output of the DL models was discretized by using the values of 10, 20, 30 and 40 for the threshold τ (corresponding to light to heavy rainfall). Table 1 presents the results obtained by evaluating the performance of the predictions generated by SepConv-ens with a lead time ranging from 6 to 48 minutes ahead. As expected, the metrics' values for the non-zero valued points are significantly worse than those computed for all data points and the performance of the classification is lower for higher τ values. These suggest an underestimation of high reflectivity values and are due to the severe imbalance of the weather data used for training, where high values are rare. We also note that the values for the evaluation measures generally decrease as the lead time increases and this is due to the fact that the prediction errors are propagated from one time step to another.

Lead time	Regression metrics				Classification metrics					Image processing metrics			
	$RMS E (\downarrow)$		$R^2 (\uparrow)$		τ	$POD (\uparrow)$	$FAR (\downarrow)$	$CSI (\uparrow)$	$HSS (\uparrow)$	$VIF (\uparrow)$		$F_{cbc} (\uparrow)$	
	All	Non-zero	All	Non-zero						All	Non-zero	All	Non-zero
6 min (1 step)	0.963	7.000	0.446	0.324	10	0.604	0.066	0.580	0.729	0.103	0.120	0.971	0.114
					20	0.439	0.076	0.424	0.593				
					40	0.249	0.386	0.215	0.354				
18 min (3 steps)	1.927	10.31	0.094	0.396	10	0.503	0.109	0.474	0.637	0.014	0.013	0.955	0.049
					20	0.336	0.122	0.321	0.484				
					40	0.082	0.572	0.074	0.137				
30 min (5 steps)	1.838	10.17	0.094	0.496	10	0.434	0.129	0.537	0.480	0.012	0.011	0.955	0.046
					20	0.249	0.129	0.240	0.384				
					40	0.040	0.528	0.038	0.074				
48 min (8 steps)	1.532	9.722	0.190	0.539	10	0.249	0.234	0.232	0.369	0.016	0.014	0.963	0.047
					20	0.084	0.275	0.082	0.149				
					40	0.006	0.875	0.006	0.011				

Table 1: Experimental results for SepConv-ens.

A visualisation of SepConv-ens predictions are shown in Figure 3, for a sample time step. One observes that the quality of the prediction degrades as the lead time increases. Interesting to note is that the quality of the prediction decreases as the elevation angle increases – the best predictions are for R01, while predictions for R02 are slightly worse. This is due to the fact that the data at higher elevation angles has fewer and fewer non-zero values, thus making it harder and harder to learn. As previously mentioned, there are much fewer data points with high values than with low values, thus making it harder to correctly learn to predict higher values. Even with the modified loss to account for this, it is clear that the model underestimates the values it predicts, especially the highest values, and this problem is more evident as the quality of the prediction degrades. We also note that the resulted values are *smoothed out*, meaning that values close together tend to be similar, and changes in the class of values tends to be much more gradual and smooth, while in the real data it can be seen that high values and low values are often adjacent and there can be sharp differences in the values in relatively small locations. Moreover, smoothing out the values means that small locations of non-zero values are not predicted while the shape of the more severe events is “filled out”, adding non-zero values in locations that should be zero, because the model cannot predict a shape as jagged as in reality.

For answering **RQ2**, we have conducted additional experiments to empirically motivate the way the three base models Bm_1, Bm_3, Bm_5 are combined for obtaining SepConv-ens's predictions up to 48 minutes ahead. Three experiments were carried out with the goal of emphasizing that SepConv-ens extrapolates the radar data (with a lead time of 18, 30 and 48 minutes respectively) better than an alternative model which uses other possible combinations of the base models. Thus, for the first experiment (**E1**) we compared how accurate is the prediction with 3 time steps ahead ($t + 3$) obtained by SepConv-ens (for which the $t + 3$ step was predicted by Bm_3) compared to the prediction of an alternative model **M1** that extrapolates the radar data at $t + 3$ by applying three times the prediction with only 1 step ahead using Bm_1 (i.e., $\widehat{M}_{t+3} = \widehat{\Phi}_1(M_{t-1}, M_t, \widehat{M}_{t+1}, \widehat{M}_{t+2})$ where $\widehat{M}_{t+1} = \widehat{\Phi}_1(M_{t-3}, M_{t-2}, M_{t-1}, M_t)$ and $\widehat{M}_{t+2} = \widehat{\Phi}_1(M_{t-2}, M_{t-1}, M_t, \widehat{M}_{t+1})$). In the second experiment (**E2**), for a target time step of 5 steps ahead ($t + 5$) we compared the prediction of SepConv-ens (for which the $t + 5$ step was predicted by Bm_1) with the prediction of an alternative model **M2** that uses Bm_3 to predict the step $t + 5$ (i.e., $\widehat{M}_{t+5} = \widehat{\Phi}_3(M_{t-1}, M_t, \widehat{M}_{t+1}, \widehat{M}_{t+2})$, where $\widehat{M}_{t+1}$ and

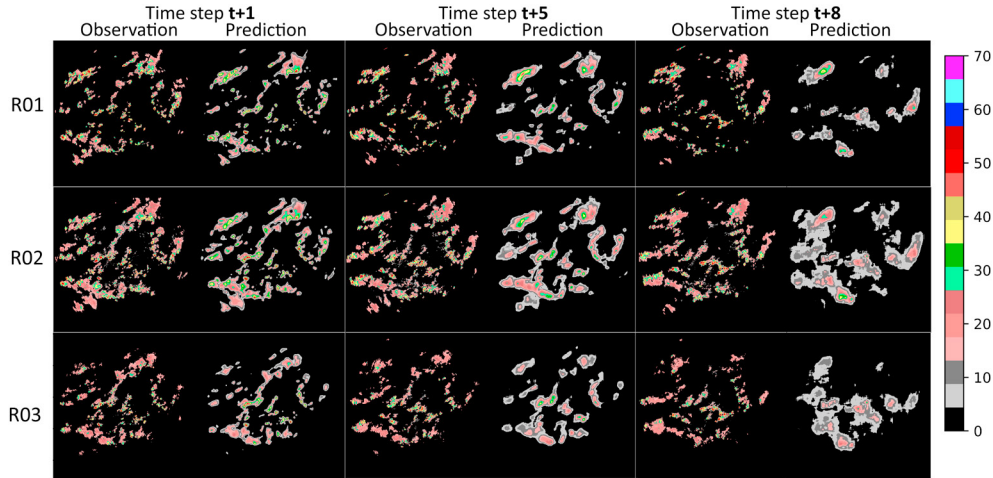


Fig. 3: A visualisation of the SepConv-ens predictions for a sample time step. The prediction shown are for the R01, R02 and R03 radar products, one on each row. The prediction for 1 step ahead (6 minutes), 5 steps ahead (30 minutes) and 8 steps ahead (48 minutes) are shown on each column.

$\widehat{M}_{t+2}$ are computed as in **E1**. For a target time step of 8 steps ahead ($t + 8$), the third experiment (**E3**) compared the prediction of SepConv-ens (for which the $t + 8$ step was predicted by Bm_5) with the prediction of an alternative model **M3** that uses Bm_3 to predict the step $t + 8$ (i.e., $\widehat{M}_{t+8} = \widehat{\Phi}_3(\widehat{M}_{t+2}, \widehat{M}_{t+3}, \widehat{M}_{t+4}, \widehat{M}_{t+5})$, where $\widehat{M}_{t+2}$, $\widehat{M}_{t+3}$, $\widehat{M}_{t+4}$ and $\widehat{M}_{t+5}$ are computed as described in Section 3.2.

Table 2 depicts the results of experiments E1-E3. For each of the conducted experiments, we present the predictive performance of SepConv-ens and of the base model(s) used for comparison (as previously described). For evaluating the results of the classification, we used the value 30 for the threshold τ , as this value is often viewed as the threshold where heavy rainfall can be considered (and heavy rainfall is often times more relevant for nowcasting). The best values for the performance metrics are highlighted in the table.

Experiment	Model	Regression metrics				Classification metrics				Image processing metrics			
		$RMSE$ ($\downarrow$)		R^2 ($\uparrow$)		POD ($\uparrow$)	FAR ($\downarrow$)	CSI ($\uparrow$)	HSS ($\uparrow$)	VIF ($\uparrow$)		F_{cbc} ($\uparrow$)	
		All	Non-zero	All	Non-zero					All	Non-zero	All	Non-zero
E1 - prediction at $t + 3$	M1	1.927	10.31	0.094	0.396	0.071	0.839	0.052	0.097	0.014	0.013	0.955	0.049
	SepConv-ens	1.149	7.874	0.353	0.674	0.245	0.293	0.222	0.363	0.054	0.058	0.968	0.086
E2 - prediction at $t + 5$	M2	1.838	10.17	0.094	0.496	0.033	0.815	0.029	0.055	0.012	0.011	0.955	0.046
	SepConv-ens	1.455	9.221	0.219	0.591	0.055	0.385	0.053	0.101	0.021	0.018	0.964	0.054
E3 - prediction at $t + 8$	M3	1.854	10.248	0.090	0.516	0.004	0.618	0.004	0.007	0.011	0.010	0.952	0.057
	SepConv-ens	1.532	9.722	0.190	0.539	0.033	0.540	0.032	0.062	0.016	0.014	0.963	0.047

Table 2: Results of Experiments E1-E3.

From Table 2 it can be seen that there is a significant difference between computing the measures for all points and computing only for non-zero points. The classification metrics are computed for all points, but CSI and HSS are measures developed explicitly for forecasting skill measurement, and they already take into account that there are fewer points in the positive class (i.e., values higher than τ) while POD and FAR are only computed for the positive class, so the imbalance of the classes does not affect the values. We can also see that the values of the VIF metric are very low, meaning that the prediction has a much worse quality than the ground truth data. This is possibly a result of the way that the prediction image is *smoothed out* by the model, a problem that is common in weather forecasting using CNNs [3]. The high difference between “All” and “Non-zero” values for the F_{cbc} measure highlights the significant data imbalance and indicates that the models (any of them) have trouble to predict the correct class for the non-zero valued points which are underrepresented (they generally underestimate the value).

For completing the answer to **RQ2**, we are further comparing the performance of the proposed SepConv-ens model with those of similar literature approaches which addressed the problem of temporal extrapolation of radar data from the Romanian NMA. We note that an exact comparison with other related work approaches cannot be provided, as both the data sets and testing methodology differs from the ones employed in the current paper.

An CNN-based model *XNow* for radar data prediction with 6 minutes ahead was introduced by Socaci et al. [10]. Experiments were performed on radar data collected from 10 days by the Romanian NMA, with a *RMSE* of 1.85 being reported on all data points and of 2.82 on non-zero valued points. The *NeXtNow* architecture [1] was further introduced as an extension of *XNow*. Experiments performed on radar data collected from 20 days by the Romanian NMA provided the following results: an average *RMSE* of 2.539 on all data points and of 8.038 on non-zero valued points; *CSI/FAR/POD* values of 0.578/0.066/0.605 for $\tau = 10$ and 0.266/0.081/0.275 for $\tau = 20$ and a lead time of 6 minutes. We note that our SepConv-ens model obtains better *RMSE* values on all data points (0.963) compared to both *XNow* and *NeXtNow*. Considering only the non-zero valued points, SepConv-ens surpasses *NeXtNow* in terms of *RMSE* but is less performant than *XNow* (this may be due to the dimensionality of the testing data - about 20 times higher for SepConv-ens). In what concerns the classification performance, SepConv-ens outperforms the *NeXtNow* model for both $\tau = 10$ and $\tau = 20$ in terms of *CSI*, *FAR* and *POD* metrics. We note that the performance of SepConv-ens is comparable with the performance provided in the literature, as shown in Section 2 (*CSI*, *POD* and *HSS* above 0.5 and *FAR* below 0.3), at least for lower classification thresholds and lead times.

As an answer to **RQ3**, SepConv-ens model was deployed in an operational environment, at the Romanian Meteorological Administration. SepConv-ens was integrated within the existing IT infrastructure and operational workflows of the Romanian NMA meteorological service. The images from Figure 4 depict a side-by-side comparison of the actual composite reflectivity, derived from data captured by the Bobohalma weather radar (shown in the left panel), and the predicted composite reflectivity generated by SepConv-ens predictive model (the right panel). The graphs are obtained using Visual Weather, the operational visualization software utilized by the Romanian Meteorological Administration. The featured event occurred on November 15, 2023, with data collected between 0:36 and 9:03 UTC during the temporal evolution of the phenomena (Figures 4.(a)-(c)) and the end of the event (Figure 4.(d)). It's worth noting that the absence of azimuth sector data is due to a radar setting designed to limit the scanning volume; this restriction is only temporary and arises from the ongoing installation of a new meteorological radar at the site. The results from Figure 4 show a good correlation between the radar measurements and SepConv-ens predictions in terms of spatial and intensity evolution of the radar echoes.

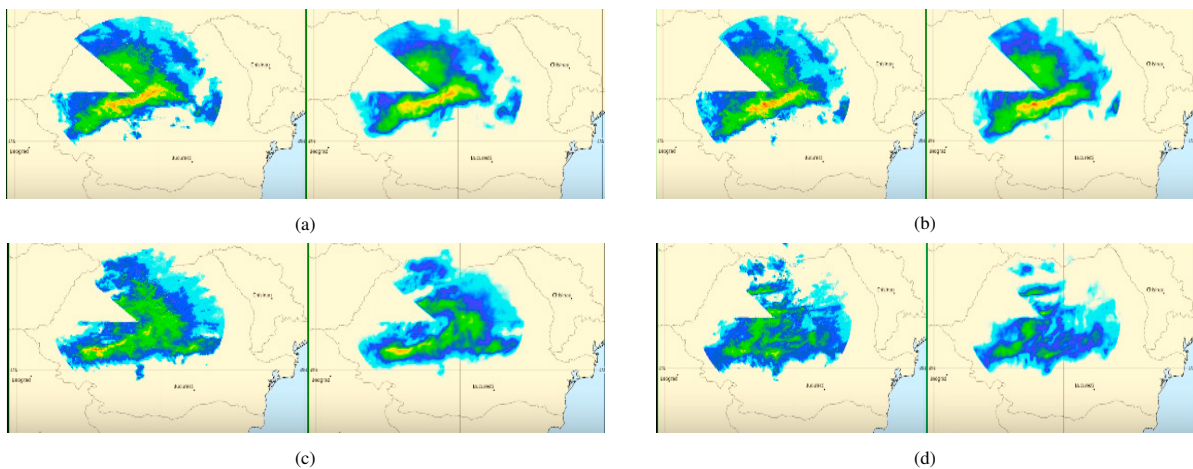


Fig. 4: The images depict a comparison of the actual composite reflectivity, derived from data captured by the Bobohalma weather radar (shown in the left panel), and the predicted composite reflectivity generated by SepConv-ens model (the right panel). .

Analysing the performance of the predictions in different weather contexts, the nowcasting meteorologists concluded that the precision of predictions generated by SepConv-ens is higher in the first 30 minutes, with a good estimation of both the spatial extent of the convective systems and of the composite reflectivity, but the accuracy decreases as the prediction time frame increases. Also, the results of the SepConv-ens forecast for stratiform rain events are better compared to convective events. It has to be noted that the chaotic character of the atmosphere makes precise weather forecasting over extended intervals a very difficult task. As the anticipation window extends, the precision of any forecasting method diminishes due to the intricate interactions within the atmosphere. This is the paradigm in the case of established methodologies such as Numerical Weather Prediction models, but also in the case of DL models.

We acknowledge that a current limitation of SepConv-ens is that it uses non-recurrent DL models, while recurrent architectures are better suited for time series forecasting by addressing the temporal nature of radar data extrapolation. However, we considered the temporal characteristic of the problem by including radar data from about 20 minutes before the current time moment as input for the prediction.

5. Conclusions and future work

The study introduced the SepConv-ens approach for temporal extrapolation of radar products' values up to more than 40 minutes ahead using an ensemble of three separable convolution-based DL models. We note the generality of SepConv-ens, as it can be extended for prediction of multiple time steps ahead by combining more than three base models. Besides the number of the base models, the method for combining them can be optimized in order to increase the performance of the prediction with more than 30 minutes in the future. Real radar data from the Romanian NMA was employed for evaluating the predictive performance of the proposed model. For validating the performance of the model in an operational environment, SepConv-ens was deployed at the Romanian NMA and integrated in the operational visualization software utilized by the operational meteorologists. The experimental results highlight the effectiveness of the model in temporal radar echo extrapolation and that it can be beneficial as a decision support tool in operational environments.

As a first direction of future work we aim to extend the architecture to a recurrent one and to explore other architectural changes such as skip connections and dilated convolutions. For enhancing the predictive performance, we may also consider to combine data from multiple sources such as radar data and satellite images. To improve the visual quality of the predicted images, the blurriness may be reduced by employing specific methods for the image processing field such as perceptual loss functions and denoising diffusion-based generative models.

References

- [1] Albu, A.I., Czibula, G., Mihai, A., Czibula, I.G., Burcea, S., Mezghani, A., 2022. NeXtNow: A Convolutional Deep Learning Model for the Prediction of Weather Radar Data for Nowcasting Purposes. *Remote Sensing* 14, ID: 3890.
- [2] Andrew Khalel, 2020. sewar 0.4.4 PyPI package. <https://pypi.org/project/sewar/>. Accessed: 2024-05-19.
- [3] Ayzel, G., Scheffer, T., Heistermann, M., 2020. Rainnet v1. 0: a convolutional neural network for radar-based precipitation nowcasting. *Geoscientific Model Development* 13, 2631–2644.
- [4] Bowler, N.E., Pierce, C.E., Seed, A., 2004. Development of a precipitation nowcasting algorithm based upon optical flow techniques. *Journal of Hydrology* 288, 74–91.
- [5] Chen, S., Shu, T., Zhao, H., Zhong, G., Chen, X., 2023. TempEE: Temporal–Spatial Parallel Transformer for Radar Echo Extrapolation Beyond Autoregression. *IEEE Transactions on Geoscience and Remote Sensing* 61, 1–14.
- [6] Klambauer, G., Unterthiner, T., Mayr, A., Hochreiter, S., 2017. Self-Normalizing Neural Networks, in: *Proceedings of NIPS*, pp. 1–10.
- [7] Niu, X., Zhang, L., Wang, C., Shen, K., Tian, W., Liao, B., 2023. A Generative Adversarial and Spatiotemporal Differential Fusion Method in Radar Echo Extrapolation. *Remote Sensing* 15, ID: 5329, 1–17.
- [8] Sheikh, H., Bovik, A., 2006. Image information and visual quality. *IEEE Transactions on Image Processing* 15, 430–444.
- [9] Shen, X., Meng, K., Zhang, L., Xianyu, Z., 2022. A method of radar echo extrapolation based on dilated convolution and attention convolution. *Scientific Reports* 12, 10572, 1–12.
- [10] Socaci, I.A., Czibula, G., Ionescu, V.S., Mihai, A., 2020. XNow: A deep learning technique for nowcasting based on radar products' values prediction, in: *Proceedings of SACI 2020*, IEEE Hungary Section. pp. 117–122.
- [11] Tian, L., Li, X., Ye, Y., Xie, P., Li, Y., 2020. A generative adversarial gated recurrent unit model for precipitation nowcasting. *IEEE Geoscience and Remote Sensing Letters* 17, 601–605.
- [12] Wang, S., Jiang, K., Huang, J., Huang, L., 2022. GMA: An Improved Framework of Radar Extrapolation Based on Spatiotemporal Sequence Neural Network. *Earth and Space Science* 9, e2022EA002502, 1–13.
- [13] Wen, Y., Zhang, J., Wang, D., Wang, C., Wang, P., 2023. Research on Radar Echo Extrapolation Method by Fusing Environment Grid Point Field Information. *Atmosphere* 14, ID: 980, 1–16.
- [14] Yin, J., Gao, Z., Han, W., 2021. Application of a radar echo extrapolation-based deep learning method in strong convection nowcasting. *Earth and Space Science* 8, e2020EA001621, 1–23.
- [15] Zhang, Y., Geng, S., Tian, W., Ma, G., Zhao, H., Xie, D., Lu, H., Lim, K.S., Kenny, T.C., 2023. Weather Radar Echo Extrapolation with Dynamic Weight Loss. *Remote Sensing* 15, ID: 3138, 1–19.
- [16] Zheng, K., Liu, Y., Zhang, J., Luo, C., Tang, S., Ruan, H., Tan, Q., Yi, Y., Ran, X., 2022. GAN-argcPredNet v1.0: a generative adversarial model for radar echo extrapolation based on convolutional recurrent units. *Geoscientific Model Development* 15, 1467–1475.
- [17] Zheng, K., Tan, Q., Ruan, H., Zhang, J., Luo, C., Tang, S., Yi, Y., Tian, Y., Cheng, J., 2023. GAN-argcPredNet v2.0: A Radar Echo Extrapolation Model based on Spatiotemporal Process Intensification. *Geoscientific Model Development Discussions* 2023, 1–25.